Simulated RFP Evaluation: Off-Road Haul Truck Tires

SIMULATED EXERCISE: all suppliers, prices, and performance data are fictitious and created for portfolio demonstration only.

Purpose	Portfolio exercise showing a competitive sourcing evaluation from criteria to award recommendation: weighted scoring plus a 5-year total-cost-of-ownership (TCO) comparison.
Scenario	A fictional aggregates producer is sourcing off-road haul truck tires (radial 29.5R25 class) for a 40-truck fleet under a 5-year supply agreement. Three fictitious suppliers bid.
Disclaimer	SIMULATED. Company names are invented; any resemblance to real companies is coincidental. All bid values and performance claims are made-up assumptions, not market data, and are not connected to any real employer or buyer.
Method	1) Define weighted criteria. 2) Collect bids on a standard pricing template. 3) Model 5-year TCO (purchase, freight, support, downtime, warranty credits, working-capital benefit), discounted to NPV. 4) Score each supplier 1 to 5 per criterion. 5) Apply weights, rank, and test sensitivity to the weights.
Sheet guide	Assumptions | Criteria | Pricing_Template | Bids | TCO | TCO_Summary | Scoring | Sensitivity | Recommendation
Legend	Blue text = hardcoded input. Black = formula. Green = link from another sheet. Yellow fill = key assumption to review. Change blue cells and every result recalculates.
Key limits	Premature failure rates, tire life, and delivery performance are treated as supplier-reported and unverified. A real evaluation would verify them with references, site trials, and warranty-claim history. Tires purchased per year are modeled as a steady average (fractional) for simplicity.

Assumptions

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Assumption	Value	Basis
Trucks in fleet	40	Invented
Tires per truck	6	Standard 6-tire haul truck layout
Operating hours per truck per year	3,000	Invented
Annual tire-hours (fleet)	720,000	Calculated: trucks x tires x hours
Downtime hours per premature tire failure	6.0	Invented
Downtime cost per hour ($)	$450	Invented: lost production and standby labour
Cost of capital / discount rate	7.0%	Invented; used for NPV and working-capital benefit
Contract term (years)	5	Model is built for 5 annual columns

Evaluation Criteria and Weights

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Criterion	Weight	What is assessed	How scored
Total cost of ownership	40.0%	5-year NPV cost: purchase, freight, support, downtime, warranty credits, working capital	Formula: 5 x lowest NPV / supplier NPV
Technical performance & quality	20.0%	Guaranteed tire life, premature failure rate, spec compliance	Evaluator score 1 to 5
Delivery & service	15.0%	Lead time, on-time delivery, onsite support	Evaluator score 1 to 5
Supplier risk & compliance	10.0%	Financial stability, supply continuity, insurance, safety compliance	Evaluator score 1 to 5
Commercial terms	10.0%	Payment terms, warranty terms, price escalation protection, flexibility	Evaluator score 1 to 5
Sustainability & safety practices	5.0%	Tire recycling or retread programs, safety record, ESG policy	Evaluator score 1 to 5
Total weight (must equal 100%)	**100.0%**	OK	

Score	Meaning
5	Excellent: clearly exceeds requirements
4	Good: meets requirements with minor strengths
3	Acceptable: meets minimum requirements
2	Weak: gaps or notable concerns
1	Poor: fails requirements or unacceptable risk

Supplier Pricing Template (issued to bidders)

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Instructions: complete one row per item. Prices in CAD, delivered to the buyer's yard, excluding GST. Bids must hold for 90 days. Blue row is an EXAMPLE of the expected format.

Line	Item description	Unit	Est. annual qty	Unit price ($)	Freight ($/unit)	Annual escalation (%)	Payment terms (days)	Guaranteed life (hours)	Warranty credit on early failure (%)	Lead time (weeks)	Notes
1	EXAMPLE: Radial 29.5R25 rock-service haul truck tire	Each	150	$4,300	$90	3.0%	30	4,400	50.0%	4	Example only
2											
3											

Supplier	declarations (bidder to confirm):
[ ]	Valid commercial liability insurance certificate attached
[ ]	Safety program and incident record attached
[ ]	Three customer references with fleet size and tire life data attached
[ ]	Warranty claim process and response time described
[ ]	All exceptions to the draft agreement listed

Bid Data Received (invented)

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Bid parameter	Unit	Northfield Tire	Summit Tread	Prairie Rubber
Unit price per tire	$	$4,200	$4,650	$3,700
Guaranteed tire life	hours	4,500	5,200	3,800
Freight per tire	$	$85	$120	$60
Annual onsite support fee	$/yr	$12,000	$8,000	-
Annual price escalation	%	3.0%	2.0%	4.5%
Premature failure rate (supplier-reported)	% of tires	4.0%	2.0%	8.0%
Warranty credit on premature failures	% of price	50.0%	75.0%	25.0%
Payment terms	days	45	30	60
Lead time	weeks	4	6	3
On-time delivery, last 12 months (supplier-reported)	%	95.0%	92.0%	96.0%

All values are invented for this simulation. Lead time and on-time delivery inform the Delivery & service score only; they do not enter the TCO model.

5-Year Total Cost of Ownership by Supplier

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Northfield Tire						
Cost line	**Year 1**	**Year 2**	**Year 3**	**Year 4**	**Year 5**	**5-Year total**
Tires purchased (avg per year)	160.0	160.0	160.0	160.0	160.0	800.0
Unit price ($)	$4,200	$4,326	$4,456	$4,589	$4,727	
Purchase cost	$672,000	$692,160	$712,925	$734,313	$756,342	$3,567,739
Freight	$13,600	$13,600	$13,600	$13,600	$13,600	$68,000
Onsite support fee	$12,000	$12,000	$12,000	$12,000	$12,000	$60,000
Premature failures (tires)	6.4	6.4	6.4	6.4	6.4	32.0
Downtime cost	$17,280	$17,280	$17,280	$17,280	$17,280	$86,400
Warranty credit	($13,440)	($13,843)	($14,258)	($14,686)	($15,127)	($71,355)
Working-capital benefit (payment terms)	($5,799)	($5,973)	($6,153)	($6,337)	($6,527)	($30,790)
Total annual cost	**$695,641**	**$715,223**	**$735,394**	**$756,169**	**$777,568**	**$3,679,994**
Discount factor	0.9346	0.8734	0.8163	0.7629	0.7130	
Present value	**$650,131**	**$624,704**	**$600,300**	**$576,878**	**$554,395**	**$3,006,408**

Summit Tread						
Cost line	**Year 1**	**Year 2**	**Year 3**	**Year 4**	**Year 5**	**5-Year total**
Tires purchased (avg per year)	138.5	138.5	138.5	138.5	138.5	692.3
Unit price ($)	$4,650	$4,743	$4,838	$4,935	$5,033	
Purchase cost	$643,846	$656,723	$669,858	$683,255	$696,920	$3,350,601
Freight	$16,615	$16,615	$16,615	$16,615	$16,615	$83,077
Onsite support fee	$8,000	$8,000	$8,000	$8,000	$8,000	$40,000
Premature failures (tires)	2.8	2.8	2.8	2.8	2.8	13.8
Downtime cost	$7,477	$7,477	$7,477	$7,477	$7,477	$37,385
Warranty credit	($9,658)	($9,851)	($10,048)	($10,249)	($10,454)	($50,259)
Working-capital benefit (payment terms)	($3,704)	($3,778)	($3,854)	($3,931)	($4,010)	($19,277)
Total annual cost	**$662,576**	**$675,186**	**$688,048**	**$701,167**	**$714,549**	**$3,441,526**
Discount factor	0.9346	0.8734	0.8163	0.7629	0.7130	

Present value	**$619,230**	**$589,734**	**$561,652**	**$534,917**	**$509,463**	**$2,814,997**

Prairie Rubber						
Cost line	**Year 1**	**Year 2**	**Year 3**	**Year 4**	**Year 5**	**5-Year total**
Tires purchased (avg per year)	189.5	189.5	189.5	189.5	189.5	947.4
Unit price ($)	$3,700	$3,867	$4,040	$4,222	$4,412	
Purchase cost	$701,053	$732,600	$765,567	$800,018	$836,018	$3,835,255
Freight	$11,368	$11,368	$11,368	$11,368	$11,368	$56,842
Onsite support fee	-	-	-	-	-	-
Premature failures (tires)	15.2	15.2	15.2	15.2	15.2	75.8
Downtime cost	$40,926	$40,926	$40,926	$40,926	$40,926	$204,632
Warranty credit	($14,021)	($14,652)	($15,311)	($16,000)	($16,720)	($76,705)
Working-capital benefit (payment terms)	($8,067)	($8,430)	($8,809)	($9,206)	($9,620)	($44,132)
Total annual cost	**$731,259**	**$761,813**	**$793,741**	**$827,106**	**$861,973**	**$3,975,892**
Discount factor	0.9346	0.8734	0.8163	0.7629	0.7130	
Present value	**$683,420**	**$665,397**	**$647,929**	**$630,995**	**$614,575**	**$3,242,316**

Working-capital benefit = purchase cost x cost of capital x payment-terms days / 365 (longer terms reduce cost). Costs are positive; credits and benefits are negative.

TCO Summary: Lowest Price vs Lowest Cost

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Supplier	Unit price ($)	5-year nominal TCO ($)	5-year NPV TCO ($)	NPV vs lowest ($)	NPV vs lowest (%)	Rank by unit price (1 = lowest)	Rank by NPV TCO (1 = lowest)	TCO score (1 to 5)
Northfield Tire	$4,200	$3,679,994	$3,006,408	$191,412	6.8%	2	2	4.68
Summit Tread	$4,650	$3,441,526	$2,814,997	-	0.0%	3	1	5.00
Prairie Rubber	$3,700	$3,975,892	$3,242,316	$427,320	15.2%	1	3	4.34

TCO score = 5 x (lowest NPV TCO / supplier NPV TCO). The lowest-cost supplier scores 5.00.

Weighted Scoring

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Criterion	Weight	Northfield Tire	Summit Tread	Prairie Rubber
Total cost of ownership	40.0%	4.68	5.00	4.34
Technical performance & quality	20.0%	4.0	5.0	3.0
Delivery & service	15.0%	4.0	3.0	4.0
Supplier risk & compliance	10.0%	4.0	4.0	3.0
Commercial terms	10.0%	3.0	4.0	3.0
Sustainability & safety practices	5.0%	3.0	4.0	2.0
Weighted score (out of 5)	**100.0%**	**4.12**	**4.45**	**3.64**
Weighted score (% of maximum)		**82.5%**	**89.0%**	**72.7%**
Rank (1 = best)		**2**	**1**	**3**

Scoring rationale (invented evaluator notes)		Northfield Tire	Summit Tread	Prairie Rubber
Technical performance & quality		Mid-range life (4,500 h), 4% failure rate	Longest life (5,200 h), lowest failure rate (2%)	Shortest life (3,800 h), 8% failure rate
Delivery & service		4-week lead time, 95% on time, paid support	6-week lead time, 92% on time; slowest delivery	3-week lead time, 96% on time; no onsite support
Supplier risk & compliance		Stable, complete insurance and safety documents	Stable, complete documents, strong references	Thin financials, incomplete safety records
Commercial terms		45-day terms, 3% escalation, 50% warranty credit	30-day terms, 2% escalation cap, 75% warranty credit	60-day terms but 4.5% escalation, 25% warranty credit
Sustainability & safety practices		Retread program offered	Retread and recycling program, strong safety record	No recycling program offered

Sensitivity: How Robust Is the Award Decision?

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Weighting scenario	TCO weight	Northfield Tire	Summit Tread	Prairie Rubber	Winner
Base case	40.0%	4.12	4.45	3.64	**Summit Tread**
Cost-focused	60.0%	4.31	4.63	3.87	**Summit Tread**
Quality-focused	25.0%	3.98	4.31	3.46	**Summit Tread**
TCO ignored (non-cost only)	0.0%	3.75	4.08	3.17	**Summit Tread**

Non-TCO criteria keep their relative weights and are rescaled so all weights sum to 100% in each scenario.

Award Recommendation (simulated)

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Item	Result
Recommended supplier (highest weighted score)	**Summit Tread**
Weighted score (out of 5)	**4.45**
Recommended supplier 5-year NPV TCO	**$2,814,997**
Lowest unit-price bidder	**Prairie Rubber**
Lowest unit-price bidder 5-year NPV TCO	**$3,242,316**
Extra 5-year NPV cost of choosing the lowest unit price	**$427,320**
Same winner in every sensitivity scenario?	**Yes**

Narrative (written for the base case as built; revisit if inputs change)	
Finding	The lowest unit price does not give the lowest cost. Prairie Rubber's cheaper tires wear out sooner and fail early more often, so downtime and replacement volume outweigh the saving. Summit Tread has the highest unit price but the lowest 5-year NPV cost.
Trade-off	Summit Tread has the slowest lead time (6 weeks) and the shortest payment terms (30 days). Negotiate a lead-time commitment with a stock-holding or consignment arrangement.
Negotiation levers	Ask Summit Tread for: 45-day payment terms; a written 2% escalation cap for all 5 years; a lead-time service level with a penalty; and warranty credit paid within 30 days of a claim.
Risks	Failure rates and tire life are supplier-reported. Mitigate with a 3-month site trial, warranty-claim reporting, and an option to re-source a share of volume if life falls below 90% of the guaranteed hours.
Next steps	Verify references, confirm insurance, finalize the agreement draft, and set renewal and expiry reminders in the contract repository.